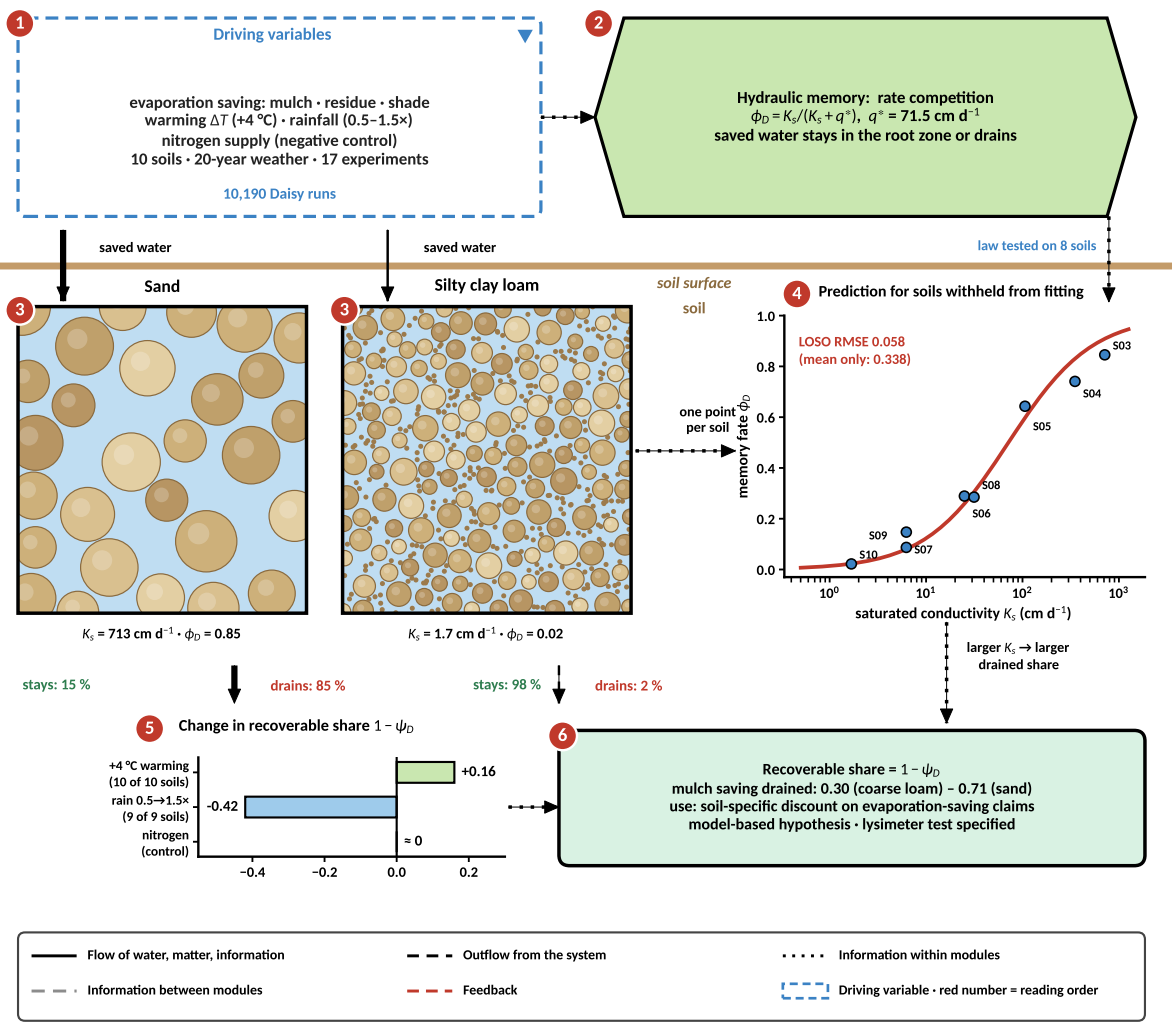

1
Driving variables
evaporation saving: mulch · residue · shade
warming ΔT (+4 °C) · rainfall (0.5–1.5×)
nitrogen supply (negative control)
10 soils · 20-year weather · 17 experiments
10,190 Daisy runs
2
Hydraulic memory: rate competition
$\phi_D = K_s/(K_s + q^*)$, q^* = 71.5 cm d^{-1}
saved water stays in the root zone or drains
law tested on 8 soils
saved water
saved water
Sand
Silty clay loam
soil surface
soil
3
3
K_s = 713 cm d^{-1} · ϕ_D = 0.85
K_s = 1.7 cm d^{-1} · ϕ_D = 0.02
one point per soil
4
Prediction for soils withheld from fitting
LOSO RMSE 0.058
(mean only: 0.338)
memory fate ϕ_D
saturated conductivity K_s (cm d^{-1})
S03
S04
S05
S08
S06
S09
S10
S07
larger K_s → larger drained share
stays: 15 %
drains: 85 %
stays: 98 %
drains: 2 %
5
Change in recoverable share 1 − ψ_D
+4 °C warming (10 of 10 soils)
rain 0.5→1.5× (9 of 9 soils)
nitrogen (control)
-0.42
+0.16
≈ 0
6
Recoverable share = 1 − ψ_D
mulch saving drained: 0.30 (coarse loam) – 0.71 (sand)
use: soil-specific discount on evaporation-saving claims
model-based hypothesis · lysimeter test specified
Flow of water, matter, information
Outflow from the system
Information within modules
Information between modules
Feedback
Driving variable · red number = reading order